Where the CHIPS Fall: Local Labor Market Impacts of Semiconductor Mega-Investments in Maricopa County, Arizona

Pinghui Wu* Annie Liu†

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Abstract

This paper examines the local labor market effects of mega-scale semiconductor investment in Maricopa County, Arizona, which has received about $300 billion in commitments from TSMC and Intel through the 2030s with support from the CHIPS and Science Act, among the largest concentrations of industrial investment in a single county in recent US history. Using synthetic difference-in-differences, we find large employment gains, but few within manufacturing. Semiconductor manufacturing employment shows no measurable net change, as hiring at the new entrant offsets contraction at the incumbent. Gains instead concentrate in construction and in supply chain sectors, mostly equipment wholesale trade and technical services, and average about 11,600 to 16,400 jobs over the post-treatment period. Federal grants cost about $136,000 to $190,000 per job-year, well above comparable estimates in the literature. We attribute these patterns to the industry's capital intensity, the netting of entrant and incumbent employment, and a scale threshold for the relocation of specialized suppliers.

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*Federal Reserve Bank of Boston

†Federal Reserve Bank of Boston (at time of research)

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1 Introduction

This paper examines the local labor market effects of mega-scale semiconductor investment, using Maricopa County, Arizona, as a case study. In the United States, dependence on foreign semiconductor supply has become a critical national security concern. To strengthen domestic competitiveness and supply chain resilience, Congress enacted the CHIPS and Science Act in August 2022, allocating $52.7 billion in direct federal funding alongside an estimated $24 billion in tax provisions for semiconductor manufacturing facilities, research and development, and workforce training (Congressional Budget Office, 2022).[1] Beyond addressing supply chain security, the act reflects a broader federal effort to improve employment opportunities through manufacturing reshoring, creating sustainable pathways to middle-skill, high-wage manufacturing jobs (Muro and Yang, 2023). The legislation culminated years of effort, building on earlier initiatives introduced in 2020 and 2021, during which federal officials also negotiated directly with major chip manufacturers to encourage domestic production (Congressional Research Service, 2023; The New York Times, 2019).

These policy initiatives have spurred large capital commitments. Since 2020, major domestic and foreign semiconductor manufacturers have announced multibillion-dollar investment plans, with individual commitments reaching $265 billion across multiple years (Table A1). This scale substantially exceeds typical manufacturing benchmarks. Indeed, a single commitment of $265 billion surpasses the entire US manufacturing sector's annual capital expenditure, which averaged $174 billion annually during the 2018–2021 period.[2] Two factors account for the magnitude of these commitments. First, the semiconductor industry is inherently capital intensive and highly concentrated, with advanced manufacturing dominated by a handful of global firms. Second, the CHIPS and Science Act provides large financial incentives: $39 billion in direct grants and loans for facility construction and equipment, complemented by a 35 percent (up from 25 percent) Advanced Manufacturing Investment Tax Credit on qualified capital expenditures (Congressional Research Service, 2023; U.S. Code, 2025).

Despite the unprecedented scale of these commitments, evidence on the act's local labor market effects remains limited. Erten et al. (2026) examine labor market impacts across the full set of counties potentially affected by the act, estimating gains per affected county of 110 to 180 jobs in the core semiconductor sector and 140 to 200 jobs in nonresidential construction. The effects of individual mega-scale investments, those exceeding tens of billions of dollars, which are the legislation's primary beneficiaries, remain understudied.

[1] "CHIPS" stands for *Creating Helpful Incentives to Produce Semiconductors*.

[2] Source: US Census Bureau Annual Survey of Manufactures and authors' calculations.

We address this gap in the literature by analyzing Maricopa County, Arizona, which offers two advantages. First, the county has received an extraordinary concentration of semiconductor investment: Taiwan Semiconductor Manufacturing Company (TSMC) is establishing new fabrication facilities (or "fabs"), and Intel is expanding existing operations. The combined commitments total about $300 billion through the 2030s, among the largest concentrations of industrial investment in a single county in recent US history and a policy-significant case in its own right. Second, implementation has progressed further in Maricopa County than elsewhere. The new fabs of TSMC and Intel began large-volume production in the fourth quarters of 2024 and 2025, respectively; most of the other announced facilities are still under construction. This advanced timeline allows us to observe labor market adjustments as mega-scale semiconductor investment moves beyond the construction phase.

Our synthetic difference-in-differences (SDID) estimates show substantial employment gains from 2021Q2, when construction began, through 2025Q4, but few of these gains occur within the manufacturing sector. Nonresidential building construction employment averaged 33 percent above its synthetic counterfactual over the post-treatment period, and building equipment contractors were 13 percent above. Together, they represented about 9,600 additional jobs. Across the eight supply chain sectors we examine, gains range from 5 to 46 percent, totaling about 7,100 jobs, more than 80 percent of which are in wholesale trade and technical services. Semiconductor manufacturing employment shows no measurable net change over the study period. These effects grew over the post-treatment period as construction advanced and the facilities moved toward production. Averaged over the post-treatment period (2021:Q2 through 2025:Q4), the total gain ranges from about 11,600 jobs in the four sectors whose estimates are statistically significant at the 5 percent level under both of our placebo-based inference methods to about 16,400 jobs across all sectors we examine, regardless of statistical significance. These estimates cover the construction phase and the first quarters of large-volume production and should be interpreted as the near-term rather than the steady-state effects.

Three features of the investment account for much of this composition. First, semiconductor fabrication requires little labor relative to capital, with capital expenditure per employee about six times the manufacturing average, so a given volume of investment generates few production jobs. Second, the direct estimate nets hiring at the new entrant against contraction at the incumbent. TSMC reports employing more than 3,000 workers at its Arizona fab as of early 2025, while Intel reduced its workforce through company-wide restructuring. Third, the industry relies on long and specialized supply chains. Suppliers relocate production only once local operating scale justifies the fixed costs, and until then inputs are imported and the associated employment accrues to distribution and technical services rather than to local manufacturing.

Relative to the public funds committed, these employment gains are small. We construct the cost per job measure counting employment gains in all sectors regardless of statistical significance, which is the lowest calculation our estimates support. On this basis, the $10.5 billion in CHIPS and Science Act grants to the Arizona projects implies about $136,000 per job-year, or $280,000 including the investment tax credit, where job-years are the average number of additional jobs over the post-treatment period multiplied by the period's length in years. Even under this least conservative calculation, these numbers are high relative to prior estimates from the literature, which range from about $3,500 to $4,900 per job-year from investment subsidies (Criscuolo et al., 2019; Lerche, 2025) to about $22,000 to $50,000 per job-year from fiscal stimulus policies (Shoag, 2013; Chodorow-Reich et al., 2012; Suárez Serrato and Wingender, 2016; Chodorow-Reich, 2019). This higher cost reflects the act's priorities, which center on national security and supply chain resilience rather than job creation.

Beyond the direct effects, we find suggestive evidence of general equilibrium local multiplier effects (Moretti, 2010, 2011). Employment in three nontradable service sectors (retail trade, accommodation and food services, and other services) rose 3 to 8 percent relative to synthetic controls, while effects in tradable sectors were neutral to modestly negative. Consumer prices and housing prices in Maricopa County grew 5 to 6 percentage points above the synthetic counterfactual. We interpret these patterns as descriptive rather than causal, since some of these sectors were also subject to pandemic-related shocks that we cannot fully distinguish from the investment effects.

Our findings relate to three strands of the literature. The most direct is the emerging evidence on the CHIPS and Science Act. Our estimates differ in both composition and magnitude from the county-average effects reported by Erten et al. (2026), to our knowledge the only other study of the act's employment effects. We find no measurable change in semiconductor manufacturing employment and construction gains many times larger than their per-county averages. Much of this difference reflects the scale of the investment, suggesting that averaging across recipient counties obscures the effects on the act's primary beneficiaries.

We also add to the literature on the local effects of large plants in two respects. The first is the composition of the supply chain response. Large plants raise local manufacturing productivity and employment (Greenstone et al., 2010; Garin and Rothbaum, 2025), and studies of supplier responses, which focus on manufacturing, find that in industries with long and complex supply chains the local response is slow and smaller than input-output models predict (Kaglic, 2014; Adams, 2016; Scavette, 2024). At the early stage of a far larger investment, we likewise find modest supplier gains in local manufacturing, but most supply chain gains accrue to wholesale trade and technical services, which shows that the supply chain response in an industry with long and complex supply chains can occur largely outside manufacturing. The second is the role of firm size and age.

Previous research has shown that investment subsidies raise employment at small firms but not at large ones (Criscuolo et al., 2019), and net job creation is higher at small firms (Neumark et al., 2011) and at young firms (Haltiwanger et al., 2013). Consistent with this evidence, we find strong hiring at the subsidized new entrant whereas total employment fell at the subsidized incumbent.

Finally, we relate our cost per job to the literature on investment subsidies and government spending. Slattery and Zidar (2020) has shown that the cost per promised job in firm-specific subsidy deals varies widely across industries and is especially high in capital-intensive industries. These figures, however, reflect deal terms rather than employment effects. Using estimated employment effects, we likewise find a high cost per job in a capital-intensive industry. Our setting also differs from the many programs in this literature that condition awards on job creation (Criscuolo et al., 2019; Slattery and Zidar, 2020). The CHIPS and Science Act does not impose such requirement (Congressional Research Service, 2023), and because it targets an industry rather than particular places (Juhász et al., 2024), it does not necessarily guarantee net job creation. In Maricopa County, our results suggest that a subsidized incumbent need not expand employment. Whether it does may depend on how the subsidy interacts with the firm's strategic priorities, the industry's competitive structure, or both.

The remainder of the paper proceeds as follows. Section 2 describes our data and empirical strategy. Section 3 presents the main results and discusses their implications. Section 4 concludes.

2 Method

2.1 Data

We use quarterly employment data from the Bureau of Labor Statistics (BLS) Quarterly Census of Employment and Wages (QCEW), spanning 2015:Q1 through 2025:Q4. The QCEW is an administrative data set derived from unemployment insurance (UI) tax records and covers approximately 95 percent of US employment. It reports employment counts at the county-industry-quarter level, with industry classifications detailed to the six-digit North American Industry Classification System (NAICS) code.[3]

Our primary analysis examines 11 sectors across private-sector establishments. Three sectors are directly affected by semiconductor investment: semiconductor and related device manufacturing (NAICS 334413), nonresidential building construction (NAICS 2362), and building equipment contractors (NAICS 2382). Eight sectors are indirectly affected through supply chain relationships: basic chemical manufacturing (NAICS 3251), other chemical product and preparation manufac-

[3]The QCEW excludes self-employed workers, proprietors, and unpaid family workers.

turing (NAICS 3259), industrial machinery manufacturing (NAICS 3332), architectural and structural metals manufacturing (NAICS 3323), professional and commercial equipment and supplies merchant wholesalers (NAICS 4234), machinery, equipment, and supplies merchant wholesalers (NAICS 4238), chemical and allied products merchant wholesalers (NAICS 4246), and architectural, engineering, and related services (NAICS 5413).

Our outcome measure is average quarterly employment, the mean of monthly employment across the three months of each quarter. It is log-transformed and indexed to 2015:Q1: $\tilde{y}_{it} = \ln(Y_{it}) - \ln(Y_{i,2015Q1})$. This indexation does not affect the estimated treatment effects, but it allows clearer visual assessment of the parallel pre-trends assumption.

The BLS suppresses data when publication could reveal information about individual establishments. Suppression is more prevalent at detailed NAICS levels and in smaller counties. We set suppressed values to missing and apply linear interpolation to fill gaps of one to three consecutive quarters, provided the total missing quarters do not exceed seven across the full sample period. This threshold retains counties with sporadic suppression while excluding those with persistent data quality issues. We interpolate about 0.2 percent of county-quarter-industry observations in our sample.

2.2 Empirical Strategy

We estimate the sector-level effects of semiconductor investment on the Maricopa County labor market using the synthetic difference-in-differences estimator of Arkhangelsky et al. (2021). SDID constructs a synthetic control group from a weighted combination of unaffected donor counties $i = 1, \ldots, N_0$, combining the parallel trends framework of difference-in-differences with the optimal weighting of synthetic control methods.

2.2.1 Investment Timeline

The treated unit is Maricopa County, Arizona. Treatment begins in 2021:Q2, corresponding to TSMC's April 2021 groundbreaking of its first fab (Fab 21), which followed the May 2020 investment announcement.[4] This yields $T_0 = 25$ pre-treatment quarters (2015:Q1 through 2021:Q1) and $T_1 = 19$ post-treatment quarters (2021:Q2 through 2025:Q4).

Our post-treatment window captures construction activity from TSMC beginning in 2021:Q2 and from Intel beginning in 2021:Q3, following Intel's March 2021 announcement of Fab 52 and Fab

[4]We define treatment as construction start (2021:Q2) rather than announcement date (2020:Q2). This choice places the pandemic onset and early recovery within the pre-treatment period, enabling the SDID weights to adjust for differential pandemic exposure across counties. To the extent that labor markets responded to the investment announcement prior to construction, our estimates may understate the total causal effect.

62. It also captures early-stage production from TSMC's Fab 21 (commencing 2024:Q4) and Intel's Fab 52 (commencing 2025:Q4). It does not, however, capture the remainder of committed investments. TSMC's and Intel's combined investment plans include twelve fabs and two advanced packaging facilities in Maricopa County through the 2030s (TSMC, 2025, 2024; Arizona Commerce Authority, 2026b; Intel, 2021a). Our estimates therefore reflect near-term labor market effects during the construction-intensive phase and initial operations rather than long-run effects of the full investment plan.

2.2.2 Donor Pool Construction

We construct the donor pool through a three-stage process. First, we apply geographic and treatment contamination exclusions: (1) Maricopa County's six neighboring Arizona counties (Yavapai, La Paz, Yuma, Pima, Pinal, and Gila), to avoid spillover effects and (2) counties receiving semiconductor investments exceeding $10 billion during the 2020–2025 period, as identified from public announcements in Table A1.

Second, we impose data quality restrictions. Counties must have average employment of at least 100 jobs in the examined sector during the pre–COVID-19 period of 2015 through 2019 to ensure meaningful sectoral activity and no more than seven quarters of suppressed data over the study period to ensure sufficient observations for reliable comparisons.

Third, from the counties satisfying these criteria, we select the 40 donors most similar to Maricopa County in absolute distance in log average employment during the 2015–2019 period in the relevant sector. This matching ensures that donors have comparable sectoral employment scales and limits the overfitting associated with excessively large donor pools.

2.2.3 Estimation Procedure

Following Arkhangelsky et al. (2021), the SDID estimator constructs a synthetic control using two sets of weights. Unit weights $\hat{\omega} = (\hat{\omega}_1, \ldots, \hat{\omega}_{N_0})$ determine the contribution of each donor county, selected to minimize pre-treatment prediction error while encouraging dispersion across donors through L2 regularization. Time weights $\hat{\lambda} = (\hat{\lambda}_1, \ldots, \hat{\lambda}_{T_0})$ identify the pre-treatment periods most predictive of post-treatment outcomes for control units, down-weighting less informative periods. Together, these weights produce a weighted two-way fixed effects difference-in-differences estimator under which parallel trends are more plausible than in standard difference-in-differences.

The estimated average treatment effect on the treated (ATT) is

$$\hat{\tau} = \left(\frac{1}{T_1} \sum_{t=T_0+1}^{T} Y_{\text{Maricopa},t} - \sum_{t=1}^{T_0} \hat{\lambda}_t Y_{\text{Maricopa},t} \right) - \sum_{i=1}^{N_0} \hat{\omega}_i \left(\frac{1}{T_1} \sum_{t=T_0+1}^{T} Y_{it} - \sum_{t=1}^{T_0} \hat{\lambda}_t Y_{it} \right), \tag{1}$$

which measures the difference between Maricopa County and its synthetic counterfactual in average post-treatment growth relative to a time-weighted pre-treatment baseline. We report $\hat{\tau}$ for each of the 11 sectors examined in Section 3.

To assess whether treatment effects evolve over time, we also report the treatment effect averaged over the final four quarters:

$$\hat{\tau}^{\text{final}} = \left(\frac{1}{4} \sum_{t=T-3}^{T} Y_{\text{Maricopa},t} - \sum_{t=1}^{T_0} \hat{\lambda}_t Y_{\text{Maricopa},t} \right) - \sum_{i=1}^{N_0} \hat{\omega}_i \left(\frac{1}{4} \sum_{t=T-3}^{T} Y_{it} - \sum_{t=1}^{T_0} \hat{\lambda}_t Y_{it} \right). \tag{2}$$

Comparing $\hat{\tau}^{\text{final}}$ with $\hat{\tau}$ indicates whether impacts are growing, stable, or declining as the investment progresses.

Because both estimates are expressed in log points, we convert them to employment levels for interpretation. We refer to

$$\bar{Y}_{\text{Maricopa}} = \exp\left(\sum_{t=1}^{T_0} \hat{\lambda}_t \ln Y_{\text{Maricopa},t} \right) \tag{3}$$

as baseline employment: Maricopa County's pre-treatment employment, averaged using the same time weights that the estimator applies. The implied level effect is $\bar{Y}_{\text{Maricopa}} \left(\exp(\hat{\tau}) - 1\right)$ and analogously for $\hat{\tau}^{\text{final}}$. Using the estimator's own time weights keeps the level conversion consistent with the ATT. Baseline employment also serves as the weight when we aggregate four-digit sector estimates to the two-digit level in Section 3.

2.2.4 Inference

With a single treated unit, resampling-based inference methods such as bootstrap and jackknife are not applicable, as they require variation across multiple treated units. We therefore base inference on the distribution of placebo effects. We reassign treatment at $T_0 + 1$ to one donor county, drop Maricopa County from the sample, and re-estimate the SDID effect for that donor, re-optimizing both the unit weights $\hat{\omega}$ and the time weights $\hat{\lambda}$. Repeating this for each donor in turn gives N_0 placebo estimates, one per county. We estimate all N_0 of them rather than a random subset, so the standard error and p-values reported below are not seed-dependent and remain stable

across estimations. Each placebo estimate is the post-treatment gap from the synthetic difference-in-differences estimator for a donor county, and the N_0 estimates together form the reference distribution for $\hat{\tau}$. We use the placebo distribution for inference in two ways, by its variance and by the rank of $\hat{\tau}$ within it.

The first follows Arkhangelsky et al. (2021) and treats the dispersion of the placebo estimates as the sampling variance of $\hat{\tau}$, which assumes homoskedasticity across counties:

$$\hat{V}_\tau = \frac{1}{N_0} \sum_{i=1}^{N_0} \left(\hat{\tau}_i^{\text{placebo}} - \bar{\tau}^{\text{placebo}} \right)^2, \tag{4}$$

where $\bar{\tau}^{\text{placebo}}$ is the mean across donors. This yields the interval $\hat{\tau} \pm 1.96\sqrt{\hat{V}_\tau}$ and the one-sided p-value $p^{\text{var}} = \Phi\left(-\hat{\tau}/\sqrt{\hat{V}_\tau}\right)$ against the hypothesis $H_1 : \tau > 0$, where Φ is the standard normal cumulative distribution function.

The second uses only the position of $\hat{\tau}$ among the placebo estimates,

$$p^{\text{rank}} = \frac{1 + \sum_{i=1}^{N_0} \mathbf{1}\{\hat{\tau}_i^{\text{placebo}} \geq \hat{\tau}\}}{N_0 + 1}, \tag{5}$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. The numerator counts the placebo estimates at least as large as $\hat{\tau}$, plus one for $\hat{\tau}$ itself. This is a one-sided permutation p-value in the spirit of Abadie et al. (2010), also against $H_1 : \tau > 0$. In Abadie et al. (2010), donor units are ranked by the ratio of post- to pre-treatment mean squared prediction error, which guards against reading a large post-treatment gap as an effect when the synthetic control tracked the unit poorly to begin with. Our concern is the shape of the placebo distribution rather than pre-treatment fit, so we rank the estimated ATT itself.

The two tests can diverge because the placebo distributions are skewed rather than symmetric. $\hat{V}_\tau$ is a sum of squared deviations, so a long tail in either direction raises it, and p^{var} treats the resulting standard error as if the distribution were symmetric. In eight of the 11 sectors the placebo estimates are left-skewed, and those furthest from the mean are generated by donor counties with sizable employment declines. These counties raise $\sqrt{\hat{V}_\tau}$ while very few placebo estimates actually exceed $\hat{\tau}$, so rejection requires a larger estimate than the placebo distribution itself implies and p^{var} is conservative. In the three right-skewed sectors the long tail lies above the mean, on the same side as the estimate being tested, and p^{var} is correspondingly less conservative. p^{rank} is not subject to this concern, since it counts only how many placebo estimates exceed $\hat{\tau}$. It cannot fall below $1/(N_0 + 1)$, however, or 0.024 with 40 donors, since the observed assignment is itself one of the $N_0 + 1$ permutations and is always counted in the numerator. The rank test therefore cannot separate the strongest estimates from the marginal ones.

Both tests are reported for every estimate. Each figure panel in Section 3 is annotated with p^{var} and p^{rank} alongside $\hat{\tau}$ and $\hat{\tau}^{\text{final}}$. Both annotated values are the one-sided tests defined above, taken against $H_1 : \tau > 0$, the direction the investment predicts.

2.2.5 Identifying Assumptions

The key identifying assumption is the absence of time-varying unit-specific confounders: Maricopa County must not experience time-varying shocks during the post-treatment period that (1) are correlated with the semiconductor investment timing, (2) affect labor market outcomes, and (3) cannot be represented as a reweighted combination of shocks experienced by donor counties. SDID removes bias from time-invariant unit-specific factors, such as industrial composition and geographic advantages, and from common time-varying shocks affecting all counties. It cannot, however, adjust for Maricopa County–specific time-varying confounders that diverge from the county's historical relationship with donor counties precisely when treatment begins.

Several features of our setting support this assumption. TSMC's and Intel's investment locations and timing choices appear driven by factors orthogonal to contemporaneous labor market shocks: proximity to existing semiconductor ecosystem partners, availability of large tracts of undeveloped land, and federal policy incentives from the CHIPS and Science Act and its antecedents (Brennan, 2024). Consistent with these factors driving investment choice, pre-treatment trends in the majority of the 11 sectors we examine track the weighted synthetic control closely in the years immediately preceding treatment (see Figures 2–4).

The primary threat to identification is COVID-19 pandemic confounding. TSMC began construction in April 2021, coinciding with the pandemic recovery. If Maricopa County's recovery differed systematically from that of donor counties for reasons unrelated to semiconductor investment, such as differential remote-work adoption or pandemic-related migration, our estimates would conflate the investment's causal effect with these confounders.

We conduct two robustness checks to assess COVID-19 confounding. First, we examine whether remote-work adoption systematically affected post-treatment employment growth in the 11 sectors we examine.[5] Using county-level data for areas unaffected by major semiconductor investments, we find no association between remote-work adoption rates and employment recovery in nearly all these sectors. In the isolated exception, the implied bias runs against the effect we estimate rather than generating it. Remote work is therefore unlikely to confound our main estimates. Remote-work adoption does correlate negatively with employment in the broader two-digit professional services sector, in which Maricopa County's higher adoption rates make it difficult to attribute employment

[5]Results are reported in Appendix C.1.

patterns to semiconductor spillovers (discussed in Section 3.3).

Second, we test whether differential population growth confounds our employment results. Faster post-pandemic population growth could mechanically increase employment in certain sectors independent of semiconductor investment. Using county-level data for areas unaffected by major semiconductor investments, we estimate the association between the change in net population growth rates and sectoral employment growth.[6] We find associations in a subset of the 11 sectors, but Maricopa County's post-treatment population growth deviated only slightly and insignificantly from its synthetic control, too little to bias the estimates in either direction. Population dynamics therefore do not credibly account for our findings.

These diagnostics cannot rule out confounding, but they narrow the range of alternative explanations. We interpret our estimates as the causal effect of semiconductor investment, conditional on the assumption that no time-varying unobservables specific to Maricopa County emerge at the treatment date beyond what the synthetic counterfactual captures.

3 Results

We organize the results into three categories. We first examine direct employment effects in semiconductor manufacturing and construction sectors, then focus on supply chain effects in manufacturing and service industries. We then present suggestive evidence of general equilibrium spillover effects.

3.1 Direct Effects

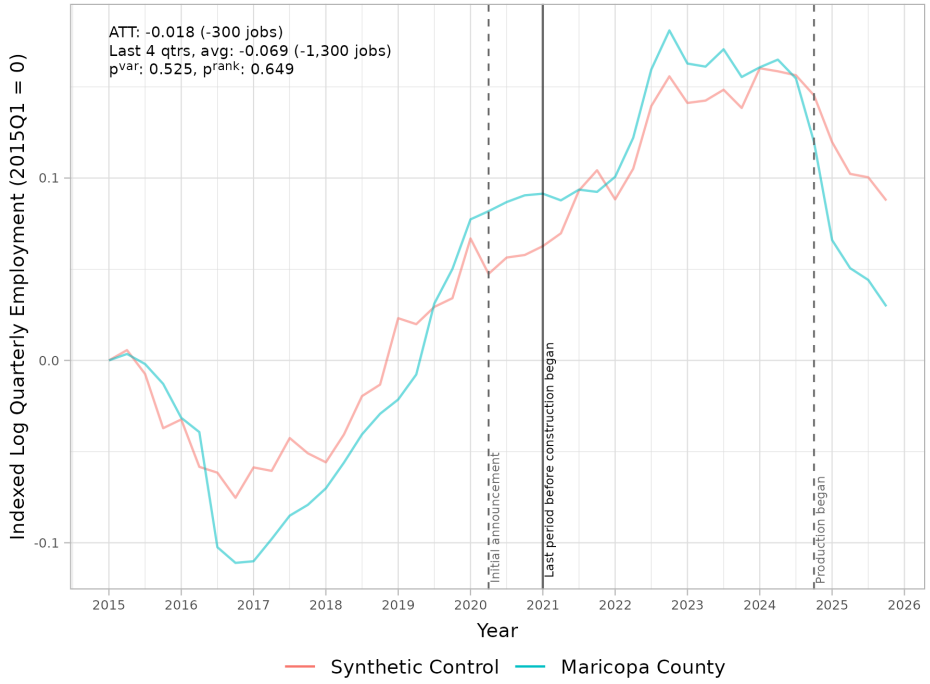
Figures 1 and 2 present synthetic difference-in-differences estimates for sectors affected directly by semiconductor investments. Each figure shows log employment trajectories for Maricopa County (blue) versus its synthetic control (red), indexed to 2015:Q1 = 0. We report two treatment effect estimates: the average over the entire post-treatment period ($\hat{\tau}$) and the average over the final four quarters ($\hat{\tau}^{\text{final}}$).

3.1.1 Semiconductor Manufacturing

Figure 1 presents results for semiconductor manufacturing (NAICS 334413). Employment in Maricopa County shows no significant divergence from its synthetic control, averaging −0.018 log points (2 percent below counterfactual) over the post-treatment period. By the 2025:Q1–2025:Q4

[6] Results are reported in Appendix C.2.

Figure 1: Estimates for Semiconductor and Related Device Manufacturing



Note(s): This figure shows the employment trajectories for NAICS 334413 from 2015:Q1 through 2025:Q4 for Maricopa County (blue) and its synthetic control (red), constructed via the synthetic difference-in-differences method. The vertical lines indicate key events in the semiconductor investment timeline. From left to right: The leftmost dashed line (2020:Q2) marks the initial investment announcement, the solid line (2021:Q1) represents the final pre-treatment period, and the rightmost dashed line (2024:Q4) marks the start of chip production at the first new fab.
Source(s): QCEW and authors' calculations.

period, observed employment had fallen 0.069 log points (7 percent) below the synthetic control, corresponding to about 1,300 fewer jobs than the counterfactual predicts.

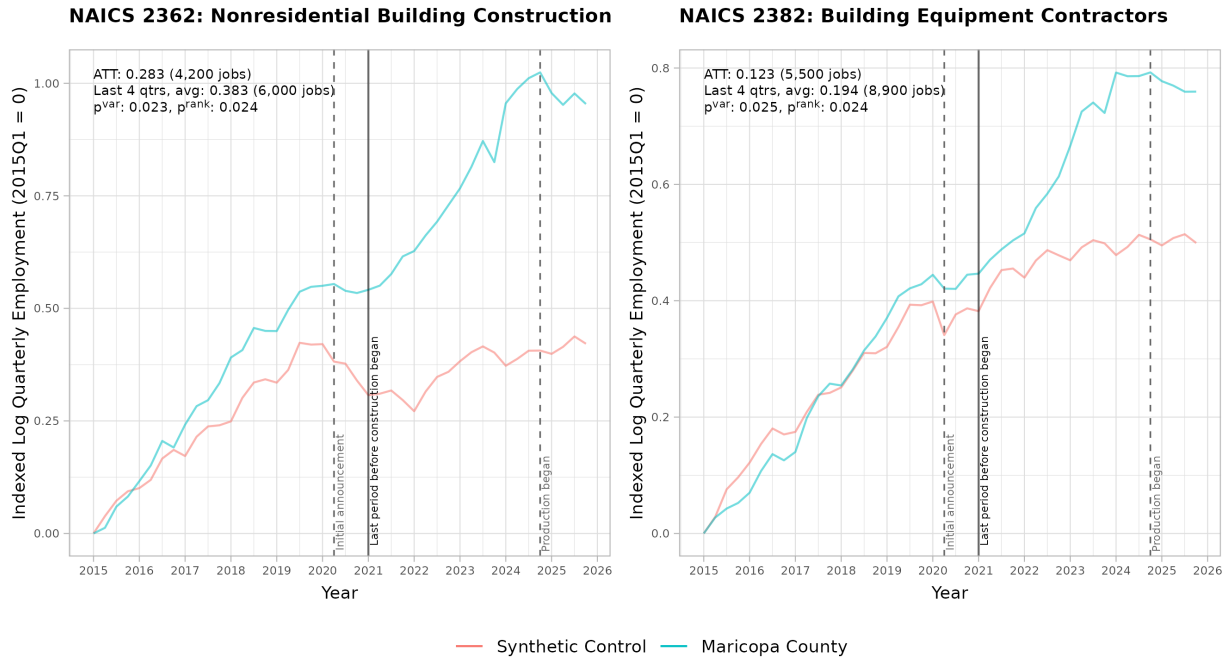
Two explanations are consistent with the absence of measurable employment gains: the sector's capital intensity and offsetting dynamics within the local industry. Semiconductor manufacturing creates few jobs per dollar invested. The 2022 US Economic Census puts capital expenditure in the industry at $100,000 per employee compared with $16,000 across all manufacturing. This ratio describes existing operations rather than the marginal effect of new investment, but a gap of that size indicates how little labor semiconductor fabrication requires relative to equipment.

The aggregate effect also masks opposing movements within the county's semiconductor sector. TSMC reported employing more than 3,000 workers at its Arizona fab as of early 2025 (TSMC, 2025), while Intel's Arizona workforce declined from 2024 to 2025 (Intel, 2024a, 2025a; AZ Job Connection, 2025). The estimate nets hiring at the new facilities against contraction at the incumbent. The Census Bureau's Quarterly Workforce Indicators (QWI), which report employment by the age of the employing firm, show this composition directly. The firm-age disaggregation is unavailable for the semiconductor industry in Maricopa County, so we measure it for Arizona, where

the county accounted for 97 percent of employment in NAICS 334413 in 2021:Q2. Beginning-of-quarter employment at firms five years old or younger rose from about 200 in 2021:Q2 to 4,100 in 2025:Q3, while employment at firms 11 years or older fell from about 21,600 to 15,100. The entrants added roughly 3,900 jobs over these quarters and the long-established firms shed about 6,500. Donor counties followed a similar path—expanding from 2021:Q2 to 2024:Q2 and then declining through 2025:Q4—over these quarters, so industry conditions likely contributed to the restructuring. A subsidy that draws new producers into a market where incumbents already operate also increases competition, which may add pressure to incumbents' margins and contribute to their contraction. Yet our design cannot establish whether the new domestic capacity added under the CHIPS and Science Act contributed to those conditions.

3.1.2 Construction

Figure 2: Estimates for Construction Sectors



Note(s): This figure shows the employment trajectories for NAICS 2362 and 2382 from 2015:Q1 through 2025:Q4 for Maricopa County (blue) and its synthetic control (red), constructed via the SDID method. The vertical lines indicate key events in the semiconductor investment timeline. From left to right: The leftmost dashed line (2020:Q2) marks the initial investment announcement, the solid line (2021:Q1) represents the final pre-treatment period, and the rightmost dashed line (2024:Q4) marks the start of chip production at the first new fab.
Source(s): QCEW and authors' calculations.

Although semiconductor manufacturing itself exhibited no net employment change during the study period, the extended construction phase of the fabs generated substantial employment in the construction sector. Figure 2 presents results for the two construction subsectors directly involved in building them: nonresidential building construction (NAICS 2362) and building equipment

contractors (NAICS 2382), which install essential systems, including electrical, plumbing, and HVAC infrastructure.

Nonresidential building construction employment in Maricopa County exceeded its synthetic counterfactual by 0.283 log points (33 percent) on average over the post-treatment period, with the effect expanding to 0.383 log points (47 percent) by the 2025:Q1–2025:Q4 period. In levels, these gains correspond to about 4,200 jobs on average and 6,000 jobs in the final quarters. Building equipment contractors exhibited a similar pattern, with employment 0.123 log points (13 percent) above the counterfactual on average and 0.194 log points (21 percent) above in the final period, corresponding to 5,500 and 8,900 additional jobs, respectively.

These magnitudes are consistent with what the two firms report for their Arizona sites. Intel describes its Arizona investment as supporting 7,000 construction jobs (Intel, 2025b), and TSMC reported that over 10,000 construction workers had worked on its Phoenix site as of December 2022 (TSMC, 2022), where construction is ongoing. The figures are not directly comparable. The firms count workers on their own sites, including those employed by establishments outside Maricopa County, while our estimates measure county employment in two construction subsectors net of what the county would have recorded absent the investments, based on the synthetic counterfactual. Nevertheless, these counts are of the same order as our estimated gains of about 9,600 construction jobs on average and 14,900 jobs in the final four quarters.

The pre-treatment trajectories for nonresidential building construction warrant further examination. The gap between Maricopa County and the synthetic control widened in 2017, before any investment activity, and then remained stable through 2020. As a robustness check, we re-estimate the sector with the pre-period beginning in 2018:Q1, so that the synthetic control is built to match Maricopa County against donor counties that were likewise growing quickly in construction before the pandemic, reported in Appendix B. The shorter window tracks the pre-treatment period far more closely, and Maricopa County's gap against the donor counties with fast pre-pandemic construction growth narrows. The ATT falls from 0.283 to 0.164 log points and the final-four-quarter average from 0.383 to 0.242. The one-sided rank p-value rises from 0.024 to 0.122. This contrast suggests that, although nonresidential building construction grew substantially in Maricopa County, the growth was less exceptional relative to the placebo counties with similarly fast pre-treatment growth. Four of the 40 donor counties experienced construction booms of their own during the post-pandemic recovery, unrelated to mega semiconductor investment, and produce placebo estimates at least as large as Maricopa County's, which raises the p-value.

3.2 Supply Chain Effects

Beyond creating semiconductor manufacturing and construction jobs, establishing fabs requires specialized materials and equipment that may generate demand in local supply chains. Materials and equipment produced locally create employment in the county's manufacturing sector, while those sourced from domestic or international suppliers generate positions in sales and technical services.

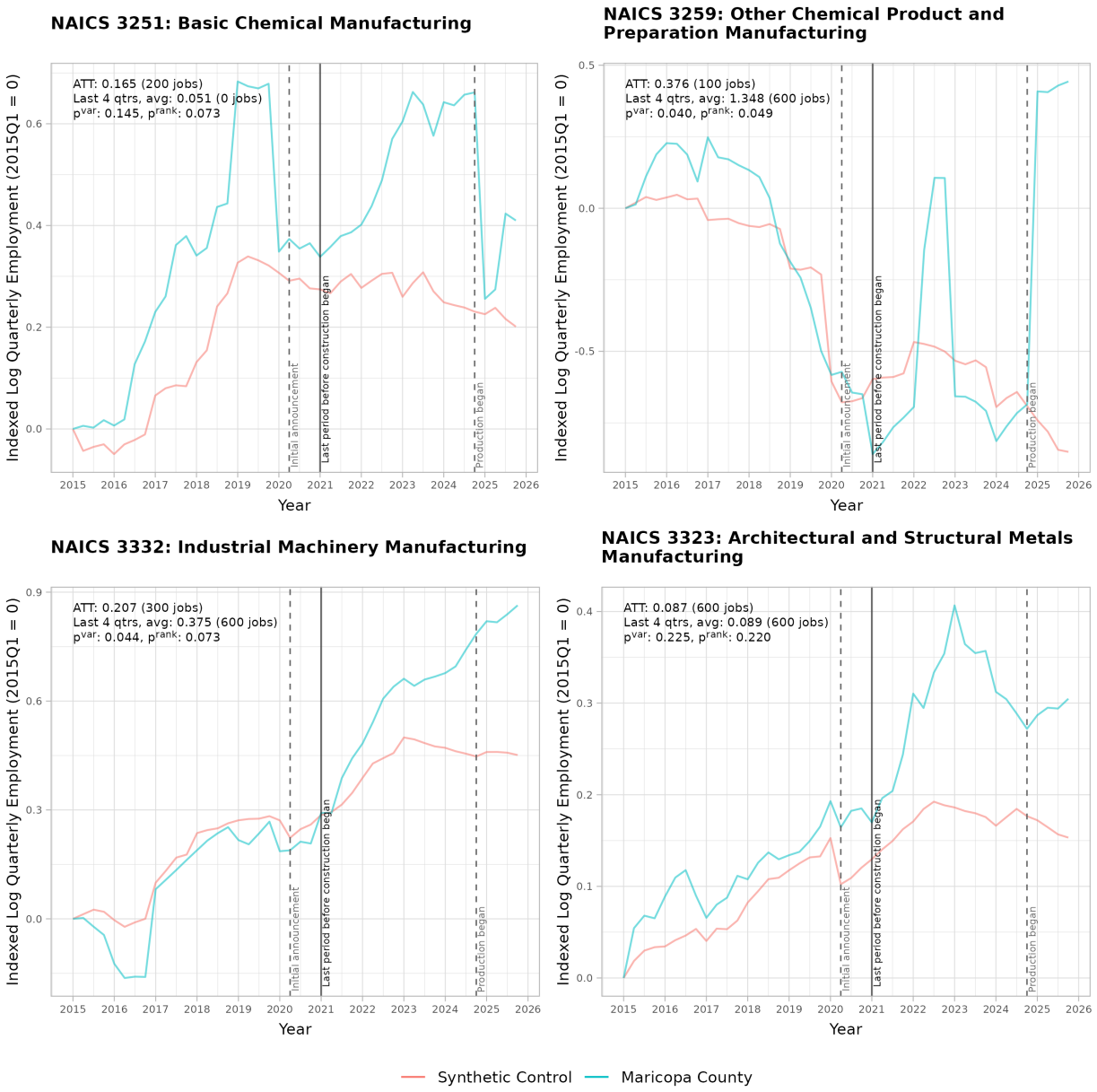
To assess these supply chain effects, Figures 3 and 4 present SDID estimates for eight sectors that serve semiconductor manufacturing demand. These sectors span both manufacturing and non-manufacturing industries, with employment outcomes shown for Maricopa County (blue) versus its synthetic control (red).

The estimated employment effects are positive in all eight sectors, consistent with expanding construction and operational activity. Figure 3 shows the four manufacturing sectors: basic chemical manufacturing (NAICS 3251), other chemical product and preparation manufacturing (NAICS 3259), industrial machinery manufacturing (NAICS 3332), and architectural and structural metals manufacturing (NAICS 3323). Figure 4 presents the four non-manufacturing sectors: professional and commercial equipment and supplies merchant wholesalers (NAICS 4234), machinery, equipment, and supplies merchant wholesalers (NAICS 4238), chemical and allied products merchant wholesalers (NAICS 4246), and architectural, engineering, and related services (NAICS 5413).

While the estimated effects are of comparable magnitude in log terms across supply chain manufacturing and service sectors, the absolute employment gains are concentrated in service industries due to their larger baseline employment levels. Employment in the four manufacturing sectors increased by 0.087 to 0.376 log points (9 to 46 percent) relative to the synthetic counterfactual in the post-treatment period, corresponding to 100 to 600 jobs per sector. Employment in the four non-manufacturing sectors grew by 0.053 to 0.231 log points (5 to 26 percent), corresponding to larger absolute gains of 500 to 2,400 jobs per sector.

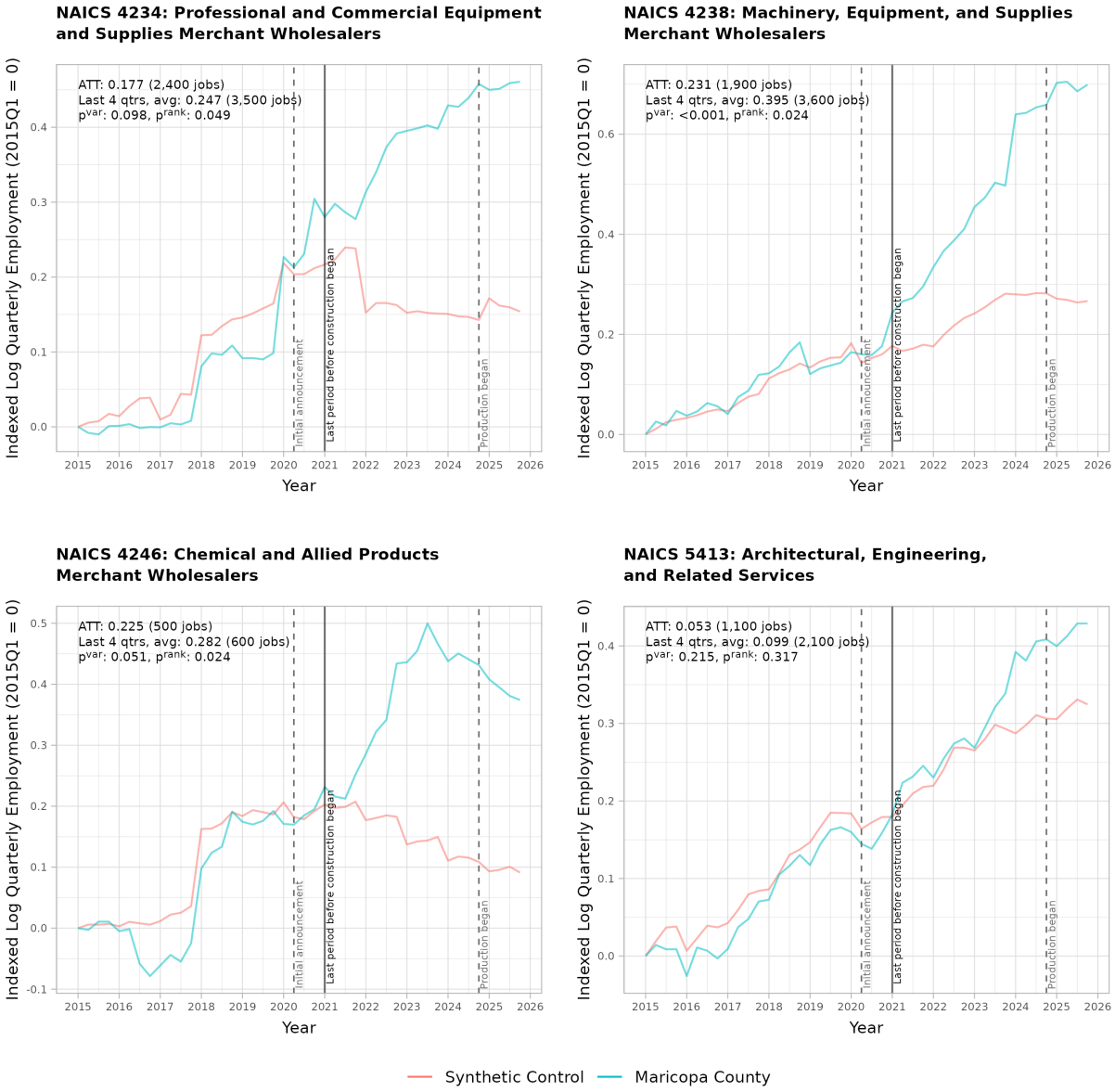
Statistical precision varies across these sectors and depends on which inference test is used. Estimates for other chemical product and preparation manufacturing (NAICS 3259) and machinery, equipment, and supplies merchant wholesalers (NAICS 4238) are statistically significant at the 5 percent level under both one-sided tests. Two other wholesale sectors, professional and commercial equipment and supplies (NAICS 4234) and chemical and allied products (NAICS 4246), are significant under the rank test but not under the variance-based test, with p^{var} of 0.098 and 0.051. Both have left-skewed placebo distributions, the case in which p^{var} is conservative, as discussed in Section 2. Industrial machinery manufacturing (NAICS 3332) shows the opposite pattern, with p^{var} of 0.044 and p^{rank} of 0.073, and its placebo distribution is right-skewed. Estimates for basic chemical

Figure 3: Estimates for Manufacturing Supply Chain Sectors



Note(s): This figure shows employment trajectories for NAICS 3251, 3259, 3332, and 3323 from 2015:Q1 through 2025:Q4 for Maricopa County (blue) and its synthetic control (red), constructed via the SDID method. The vertical lines indicate key events in the semiconductor investment timeline. The leftmost dashed line (2020:Q2) marks the initial investment announcement, the solid line (2021:Q1) represents the final pre-treatment period, and the rightmost dashed line (2024:Q4) marks the start of chip production at the first new fab.
Source(s): QCEW and authors' calculations.

Figure 4: Estimates for Non-manufacturing Supply Chain Sectors



Note(s): This figure shows employment trajectories for NAICS 4234, 4238, 4246, and 5413 from 2015:Q1 through 2025:Q4 for Maricopa County (blue) and its synthetic control (red), constructed via the SDID method. The vertical lines indicate key events in the semiconductor investment timeline. The leftmost dashed line (2020:Q2) marks the initial investment announcement, the solid line (2021:Q1) represents the final pre-treatment period, and the rightmost dashed line (2024:Q4) marks the start of chip production at the first new fab.
Source(s): QCEW and authors' calculations.

manufacturing (NAICS 3251), architectural and structural metals manufacturing (NAICS 3323), and architectural, engineering, and related services (NAICS 5413) are not statistically significant at the 5 percent level under either test.

Trade data support this service-intensive supply chain structure. Arizona's imports from Taiwan in three broad manufacturing categories—chemicals (NAICS 325), fabricated metal products (NAICS 332), and machinery (NAICS 333)—rose relative to other US states following the 2021 investment, reflecting TSMC's established supplier relationships in Taiwan (see Appendix D for import comparisons across states). Imports from a single trading partner capture only a portion of the inputs sourced for the two investments, so we treat this as partial evidence. The pattern is nonetheless consistent with the absolute employment gains accruing predominantly to wholesale distribution and technical services rather than to local manufacturing production.

3.3 General Equilibrium Spillover Effects: Suggestive Evidence

The preceding estimates imply employment gains from fab construction and operation that grew over the post-treatment period. Averaged over the post-treatment period, the gains range from about 11,600 jobs, counting only the four sectors whose estimates are statistically significant at the 5 percent level under both one-sided tests, to about 16,400 jobs across all 11 sectors regardless of statistical significance. The average gains of 11,600 to 16,400 jobs represent about 0.7 to 0.9 percent of Maricopa County's average private-sector employment from 2015 through 2019.

A general equilibrium framework predicts that employment shocks of this magnitude propagate through the local economy along two channels (Moretti, 2010, 2011). Consumption from newly added, higher-paying jobs raises demand for local nontradable services, which generates additional employment. At the same time, higher demand for labor and land raises local costs, which can crowd out employment in tradable sectors without direct investment linkages, since these firms face nationally determined output prices and cannot pass through higher input costs.

Identifying spillover effects on broader sectors, however, presents challenges distinct from those in the preceding analysis. While our diagnostics find no evidence that COVID-19–related confounding biases the construction and specialized supply chain estimates upward (see Appendix C), selected service sectors experienced concurrent post-pandemic shocks that confound identification. Remote-work adoption is one such shock. Maricopa County demonstrated above-average remote-work adoption during the post-pandemic period, and our auxiliary analysis finds that counties with higher remote-work shares experienced slower local employment growth in information (NAICS 51), finance (NAICS 52), and professional services (NAICS 54), sectors in which remote arrangements are more feasible and in which workers likely face demand competition from high-wage

remote employers. Because finance and professional services carry much of the evidence for the crowding-out channel, their negative trajectories in Maricopa County cannot be credibly attributed to semiconductor-related crowding out alone. We therefore interpret the spillover estimates, particularly those for the crowding-out channel, as suggestive.

Table 1: Employment Effects by Two-digit NAICS Sector (Excluding Semiconductor-related Employment)

Sector (NAICS code)	Weighted ATT
Non-tradable:	
Retail Trade (44–45)	0.052
Accommodation and Food Services (72)	0.032
Other Services (81)	0.073
Tradable:	
Manufacturing (31-33)	0.021
Wholesale Trade (42)	0.044
Information (51)	0.007
Finance and Insurance (52)	-0.038
Professional, Scientific, and Technical Services (54)	-0.058

Note(s): Two-digit NAICS sectors exclude semiconductor-related employment (3344, 2362, 2382, 3251, 3259, 3323, 3332, 4234, 4238, 4246, 5413). For manufacturing (31-33), wholesale (42), finance (52), professional services (54), accommodation/food (72), and other services (81), we report weighted averages of four-digit sector effects. For retail trade (44-45) and information (51), we report direct two-digit estimates.

Table 1 reports treatment effects on employment for major two-digit NAICS sectors. These estimates are weighted averages of SDID effects across four-digit subsectors within each sector, in which weights correspond to pre-treatment baseline employment levels.[7] We exclude subsectors affected directly by semiconductor investment or its supply chain linkages to isolate spillover effects.

Employment effects are positive in the locally oriented nontradable service sectors. Retail trade employment increased 0.052 log points (5 percent) above the synthetic control, while accommodation and food services grew 0.032 log points (3 percent) and other services increased 0.073 log points (8 percent). By contrast, tradable sectors generally exhibited comparable or modestly slower growth relative to the counterfactual, with most effects ranging from −0.058 to 0.021 log points (–6 to 2 percent) across the post-treatment period.

Wholesale trade is an exception, with employment growth 0.044 log points (4 percent) above the synthetic control. The positive contributions come from subsectors serving local demand: household appliances and electronics wholesalers (NAICS 4236); hardware, plumbing, and heating equipment wholesalers (NAICS 4237); furniture and home furnishings wholesalers (NAICS 4232);

[7]For the retail trade sector (NAICS 44–45) and information sector (NAICS 51), the estimates represent SDID effects estimated directly at the two-digit level, as changes in four-digit classification schemes over time prevent subsector-level analysis.

and grocery and related products wholesalers (NAICS 4244). Hence, the difference primarily reflects the sector's more localized market orientation.

This divergence between nontradable and tradable sectors is consistent with the two-channel prediction outlined earlier, and local price indexes provide corroborating evidence. As shown in Appendix E, Maricopa County's indexed consumer price level (CPI-U) increased about 5 percentage points above the synthetic control during the post-treatment period, while the FHFA house price index rose 6 percentage points more. Although these price increases cannot be attributed exclusively to semiconductor investment, they are consistent with theoretical predictions that positive labor demand shocks elevate local prices. Such cost increases may erode Maricopa County's historical cost advantages, offering one explanation for the flat to negative tradable-sector estimates reported earlier.

3.4 Discussion

The employment gains from semiconductor investment in Maricopa County are substantial in absolute terms. These gains, however, lie largely outside manufacturing. Construction accounts for the largest share of the total, and supply chain gains are concentrated in wholesale trade and technical services. Direct semiconductor manufacturing employment shows no net change over the period we observe. Four of the 11 sectors have estimates that are statistically significant at the 5 percent level under both one-sided tests of Section 2: nonresidential building construction (NAICS 2362), building equipment contractors (NAICS 2382), other chemical product and preparation manufacturing (NAICS 3259), and machinery, equipment, and supplies merchant wholesalers (NAICS 4238). Summed across these four sectors, the average post-treatment effect is about 11,600 jobs, although the nonresidential building construction estimate is sensitive to the pre-treatment window, as discussed above. Summed across all 11 sectors regardless of statistical significance, including the negative estimate for semiconductor manufacturing, it is about 16,400 jobs.

Even so, the gains are small relative to the public funds committed to the investments. The US federal government committed $10.5 billion in direct grants to TSMC's and Intel's Arizona projects through the CHIPS and Science Act. Of this, $6.6 billion went to TSMC and $3.94 billion was the Arizona portion of Intel's $7.86 billion award (NIST, 2026e,a). The Advanced Manufacturing Investment Tax Credit adds about $11.2 billion more.[8] We divide these amounts by two measures of employment. The first is job-years, the average post-treatment effect summed across the 11 sectors and multiplied by the 4.75-year length of the post-treatment period, about 77,700 job-years. The

[8]The tax credit figure applies the 35 percent Advanced Manufacturing Investment Tax Credit to the capital expenditures for the first fabs, TSMC's Fab 21 ($12 billion announced in 2020) and Intel's Fab 52 and Fab 62 ($20 billion announced in 2021).

second is jobs created by the end of the study period, the effect in the final four quarters summed across the same sectors, about 25,300 jobs. Because both measures include every sector regardless of statistical significance, the resulting ratios are the least conservative our estimates support. They come to about $136,000 per job-year counting grants alone and $280,000 counting grants and tax credits, or $417,000 and $861,000 per job created.

Table 2: Estimated cost per job: investment subsidies and government spending

Author (year)	**Policy**	**Basis**	**Cost per job**
Panel A. This paper			
This paper	CHIPS Act, Maricopa County	Job-year	$136,000–$280,000
	CHIPS Act, Maricopa County	Job created	$417,000–$861,000
Panel B. Investment subsidies			
Freedman et al. (2023)	California Competes Tax Credit	Job created	$3,315
Criscuolo et al. (2019)	UK Regional Selective Assistance	Job-year	$3,541
Lerche (2025)	German investment tax credit	Job-year	$4,900
Garrett et al. (2020)	US bonus depreciation	Job created	$20,000–$50,000
Scavette (2024)	Boeing plant, South Carolina	Job created	$133,333
Panel C. Fiscal spending			
Shoag (2013)	US state government spending	Job-year	$22,000
Chodorow-Reich et al. (2012)	ARRA state fiscal relief	Job-year	$26,000
Suárez Serrato and Wingender (2016)	US federal spending on local areas	Job-year	$30,000
Chodorow-Reich (2019)	ARRA state fiscal relief	Job-year	$50,000
Wilson (2012)	ARRA stimulus spending	Job created	$125,000

Notes: The Basis column records what the denominator counts. A cost per job-year divides spending by the employment it supports over time, either total spending by the employment effect summed over the period or annual spending by the annual employment effect. A cost per job created divides spending by the employment effect at a single point in time. Figures on different bases are not directly comparable. All numbers are in the dollar value of the original study and are not deflated.

Table 2 compares these ratios with estimates from two literatures. Evaluations of investment subsidies, many of which condition awards on job creation in a way the CHIPS and Science Act does not (Criscuolo et al., 2019; Freedman et al., 2023; Congressional Research Service, 2023), report the lowest ratios. Measured per job-year, the estimates are about $3,500 to $4,900 (Criscuolo et al., 2019; Lerche, 2025). Measured per job created, they are about $3,300 (Freedman et al., 2023) and $20,000 to $50,000 (Garrett et al., 2020). The subsidy package for Boeing's assembly plant in South Carolina, a single firm-specific deal, is an exception at about $133,000 per job created, though this figure counts aerospace employment alone and excludes effects in other industries

(Scavette, 2024). Because we find no net employment gain within semiconductor manufacturing, a comparable within-industry ratio is not defined for our estimates. Estimates of government spending multipliers are higher, about $22,000 to $50,000 per job-year (Shoag, 2013; Chodorow-Reich et al., 2012; Suárez Serrato and Wingender, 2016; Chodorow-Reich, 2019) and $125,000 per job created over the first year of the ARRA (Wilson, 2012). Even counting grants alone, our cost per job-year and cost per job created exceed the upper ends of these ranges by a wide margin.

3.4.1 Explaining the Composition and Cost of the Gains

Capital intensity of the targeted industry. Semiconductor fabrication requires little labor relative to capital. In the 2022 Economic Census, capital expenditure per employee in the industry is about six times the average across all manufacturing. Using jobs promised in firm-specific subsidy deals in the United States, Slattery and Zidar (2020) find that the cost per promised job varies widely with the industry subsidized, from about $35,000 in financial activities to more than $900,000 in basic chemical manufacturing, which also has one of the highest ratios of capital expenditure to employment among manufacturing industries in the 2022 Economic Census. Our estimates, which measure employment effects rather than promised jobs, likewise imply a high cost per job in a capital-intensive industry. A high cost per job therefore reflects in part the capital intensity of the industry a program targets.

New entrants and incumbent producers. As the QWI firm-age evidence above shows, the direct semiconductor estimate nets hiring at the entrant against contraction at the incumbent. Two mechanisms can reduce this net effect. First, an industry-based subsidy that draws a new producer into a market where incumbents already operate increases competition, which may put pressure on incumbents' margins. Second, firms of different size and age respond differently to subsidies and grow at different rates. Investment subsidies raise employment at small firms but have no detectable effect at large firms (Criscuolo et al., 2019). Net job creation is higher at young firms, which are typically small (Haltiwanger et al., 2013; Neumark et al., 2011). Both mechanisms lower the net direct effect that enters the cost-per-job calculation.

Supply chain length. The supply chain gains in Maricopa County accrue mainly to wholesale trade and technical services rather than to local manufacturing. Semiconductor fabrication relies on long and specialized supply chains, and many of its inputs are produced abroad, as the rise in Arizona's imports from Taiwan indicates. As long as inputs are imported, local employment grows in distribution and technical services rather than production. Producing inputs locally requires suppliers to relocate, which they do only once local demand justifies the fixed costs, so a long and specialized supply chain can delay local production. Evidence from other industries with

long supply chains is consistent with this pattern. Supplier responses to large aerospace and motor vehicle assembly plants have been slow, have dispersed beyond the host metropolitan area, and have been smaller than input-output models predict (Kaglic, 2014; Adams, 2016; Scavette, 2024). Over the period we observe, this structure limits the local manufacturing employment that the investment generates.

3.4.2 Effects Beyond the Study Period

Our post-treatment period runs from 2021:Q2 through 2025:Q4 and ends shortly after TSMC and Intel began production at their first new fabs, when less than one-quarter of the $300 billion commitment had been realized. The estimates therefore describe effects during the construction phase and the first quarters of production, and they should not be interpreted as steady-state effects of the full investment. Direct manufacturing employment at the new entrant should increase as its remaining fabs begin production, although the net direct effect will also depend on employment at the incumbent's existing establishments.

Construction employment gains should decline once building is complete. TSMC's and Intel's first new fabs in Maricopa County took about three to four years from the start of construction to production, which suggests that construction employment at a project of one or two fabs declines after a similar period. The full commitment in Maricopa County, however, includes twelve fabs and two packaging facilities scheduled through the 2030s, so construction employment gains are likely to persist for the next decade.

Manufacturing employment in the supply chain may also increase if local operating scale passes the threshold at which specialized suppliers relocate production. The Arizona Commerce Authority, the state's economic development agency, has announced several semiconductor suppliers committing to production facilities in the Greater Phoenix area during our sample period (Arizona Commerce Authority, 2024a, 2026a, 2024b, 2025), though these were still at an early stage through 2025:Q4, too small to register in our estimates. Both the magnitude and the composition of the effects may therefore change as the investment proceeds, and the cost per job should decline, although it will likely remain above the benchmarks.[9]

[9]We find little evidence of spillover employment effects in the counties bordering Maricopa. Applying the same estimator to each in turn, the only estimate distinguishable from zero at the 5 percent level under both one-sided tests is building equipment contractors (NAICS 2382) in Pinal County. Three wholesale trade estimates in Pinal and Pima Counties are positive and sizable, ranging from 0.30 to 0.55 log points, and each passes one of the two tests at that level. Appendix F reports the estimates.

4 Conclusion

This paper estimates the local labor market effects of mega-scale semiconductor investment, using Maricopa County, Arizona, as a case study. Applying the synthetic difference-in-differences method, we find that employment gains are concentrated outside manufacturing. Semiconductor manufacturing employment shows no measurable net change. Construction employment exceeds its synthetic counterfactual by 13 to 33 percent over the post-treatment period, and gains across supply chain sectors range from 5 to 46 percent. Supply chain gains accrue mostly to wholesale trade and technical services, and gains in supply chain manufacturing are modest in absolute terms. Averaged over the post-treatment period, the total gain ranges from about 11,600 jobs in the four sectors whose estimates are statistically significant at the 5 percent level under both tests to about 16,400 jobs across all sectors we examine.

Three features of the investment account for much of this composition. Semiconductor fabrication requires little labor relative to capital, which limits direct employment per dollar invested. The direct estimate also nets hiring at the new entrant against contraction at the incumbent over the same period. Finally, the industry relies on long and specialized supply chains. Until local operating scale justifies the fixed costs of supplier relocation, inputs are imported and the associated employment accrues to distribution and technical services rather than to local manufacturing. The same features also make the gains small relative to the public funds committed. Counting all sectors regardless of statistical significance, CHIPS and Science Act grants to the Arizona projects imply about $136,000 per job-year, or $280,000 including the investment tax credit, well above benchmarks from the literature on investment subsidies and government spending. These magnitudes reflect the act's policy priorities, which center on national security and supply chain resilience, with job creation a co-benefit rather than the principal objective.

Our estimates cover the construction phase and the first quarters of production, when less than one-quarter of the announced investment had been realized, and they should not be interpreted as steady-state effects. Given the scale of the commitment and its timeline through the 2030s, construction employment gains are likely to persist for the next decade. Manufacturing employment may rise as the remaining fabs enter production and as greater local operating scale makes supplier relocation viable. For communities and policymakers evaluating similar investments, our estimates help calibrate expectations for how much direct manufacturing employment such projects generate, how long those gains take to appear, and how much public funding each job requires.

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Appendix

A Major Semiconductor Investment Announcements since 2020

Table A1: Major Semiconductor Manufacturing Investments in the United States

Company	County	State	Initial Amount	Current Amount	CHIPS Grants	Construction Start
TSMC	Maricopa	AZ	$12B (2020)	$265B (2026)	$6.6B	2021Q2
Intel	Maricopa	AZ	$20B (2021)	$32B (2024)	$7.86B	2021Q3
	Franklin	OH	$20B (2022)	$28B (2024)		2022Q3
Samsung	Williamson	TX	$17B (2021)	$40B* (2024)	$4.745B	2022Q1
Micron	Onondaga	NY	$20B (2022)	$250B** (2026)	$6.44B	2026Q1
	Ada	ID	$15B (2022)	$250B** (2026)		2022Q3
Texas Instruments	Grayson	TX	$30B (2021)	$40B (2025)	$1.61B	2022Q2
	Utah	UT	$11B (2023)	$11B (2023)		2023Q4
GlobalFoundries	Saratoga	NY	$11.6B (2024)	$16B (2025)	$1.50B	2025

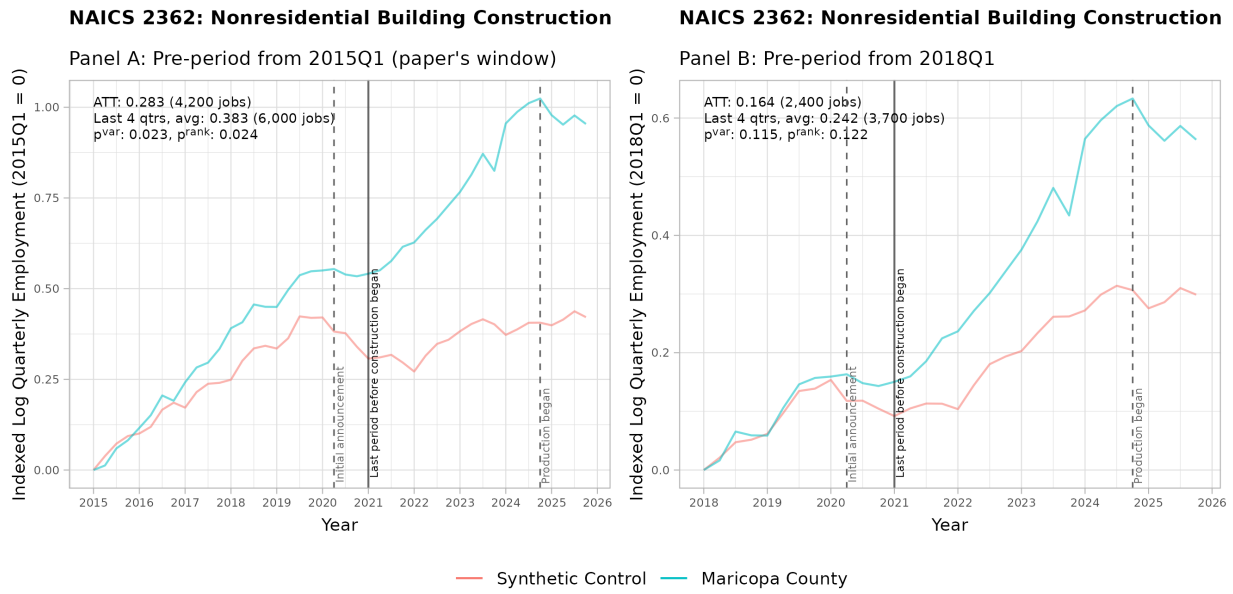
Note(s): Data compiled from public announcements and CHIPS Act awards. Initial Amount reflects the first publicly announced investment amount; Current Amount shows the commitment to date, including revisions over time. Investment announcements, CHIPS grants and loans, and construction start sources: TSMC (TSMC, 2020; U.S. Department of Commerce, 2026; NIST, 2026e; TSMC, 2026). Intel (Intel, 2021b, 2025b; Intel, 2022; Intel, 2024b, 2022; NIST, 2026a); Samsung (Office of the Texas Governor, 2021a; US Department of Commerce, 2024; NIST, 2026c; Samsung, 2026); Micron (Micron, 2022; Micron, 2026a,b; Idaho Commerce, 2024; Micron, 2022; NIST, 2026b); Texas Instruments (Office of the Texas Governor, 2021b; Texas Instruments, 2022, 2025, 2026, 2023; NIST, 2026d); GlobalFoundries (Office of the New York Governor, 2024; GlobalFoundries, 2025a, 2024, 2025b). CHIPS grants and loans represent federal funding allocated under the CHIPS and Science Act of 2022. *$40 billion allocated to the construction of a new semiconductor site in Williamson with an unspecified portion allocated to the expansion of Samsung's Austin, Texas Fab (US Department of Commerce, 2024). **$250 billion will be distributed across Micron's semiconductors sites in currently unspecified shares.

B Pre-Treatment Window for Nonresidential Building Construction

Figure A1 re-estimates NAICS 2362 with the pre-period beginning in 2018:Q1 instead of 2015:Q1, holding the donor pool, the treatment date, the outcome, and the inference procedure of Section 2 fixed. Panel A reproduces the paper's window and Panel B shows the shorter one. Each panel indexes both series to the first quarter of its own window.

The 2018:Q1 window excludes the quarters over which Maricopa County pulled ahead of its synthetic control, and the pre-treatment fit improves as a result. The root mean squared pre-treatment error falls from 0.068 to 0.019 log points.[10] The ATT falls from 0.283 to 0.164 log points and the average over the final four quarters from 0.383 to 0.242. Both one-sided p-values rise, p^{var} from 0.023 to 0.115 and p^{rank} from 0.024 to 0.122. In donor count terms, none of the 40 donor counties produces a placebo estimate as large as Maricopa's under the paper's window, while four do under the 2018:Q1 window. The same exercise for building equipment contractors (NAICS 2382) leaves that estimate almost unchanged, 0.123 against 0.117 log points, with p^{rank} at 0.024 in both windows.

Figure A1: Pre-Treatment Window Sensitivity, Nonresidential Building Construction



Note(s): Both panels estimate the same sector, NAICS 2362, with the same donor pool of 40 counties, the same treatment date of 2021:Q2, and log average quarterly employment as the outcome. They differ only in the first quarter of the estimation window: 2015:Q1 in Panel A, the window used throughout the paper, and 2018:Q1 in Panel B. Each panel indexes both series to the first quarter of its own window. The annotated p-values are the one-sided tests of Section 2. The vertical lines indicate key events in the semiconductor investment timeline.
Source(s): QCEW and authors' calculations.

[10] This error is computed after removing the average level difference between Maricopa County and its synthetic control, since the estimator differences out that level gap.

C COVID-19 Confounding: Remote Work and Population Growth

Table A2 presents regression results testing whether differential remote-work adoption or population growth during the post-treatment period confounds our main employment estimates. Both regression samples exclude counties that received major semiconductor investments (those exceeding $10 billion during the 2020–2025 period) and Maricopa County's six neighboring counties (Yavapai, La Paz, Yuma, Pima, Pinal, and Gila), and each imposes a further restriction on county size described later. We regress quarterly changes in log employment on the interaction between the post-treatment period (2021:Q2 onward) and either (i) the average remote-work adoption rate during the 2022–2025 period or (ii) the change in the county's average net population growth rate between the pre-treatment (2015–2020) and post-treatment (2021–2025) periods.

The regression specification takes the form

$$\Delta \ln(\text{employment})_{it} = \alpha_i + \gamma_t + \beta_1 \text{Confounder}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Confounder}_i \times \text{Post}_t) + \epsilon_{it} \quad \text{(A1)}$$

where α_i and γ_t are county and time fixed effects, respectively. The interaction coefficient β_3 measures whether counties with higher remote-work adoption or a larger change in population growth experienced systematically different employment trajectories in the post-treatment period. A significant coefficient would suggest that Maricopa County's observed employment patterns could reflect these confounders rather than semiconductor investment effects.

Results are reported separately for two sample restrictions: counties with 2015–2019 average population exceeding 200,000 (Sample 1) and the 40 counties closest to Maricopa County by log population distance, mirroring our SDID donor pool construction (Sample 2). Remote-work adoption rates are measured using the Current Population Survey, calculated as the share of workers reporting 35 or more hours of telework per week during October 2022 through December 2025. The population growth confounder is the change in each county's average net population growth rate, computed from the US Census Bureau's annual county population estimates, between the 2015–2020 and 2021–2025 periods.

C.1 Remote-work Adoption and Employment Growth

The remote-work results (columns 1–2) indicate minimal association between remote-work adoption and employment growth in the 11 sectors examined in our main analysis. Ten of the 11 show no statistically significant interaction in either sample. The exception is machinery and equipment wholesalers (NAICS 4238), in which the interaction is negative and significant in both samples, with coefficients of -0.048 and -0.120. Because Maricopa County's remote-work adoption rate exceeds that of comparable counties by about 2 percentage points, a negative interaction implies

Table A2: COVID-19 Confounding: Remote Work and Population Growth Robustness Checks

	Remote-Work Adoption		Population Growth	
NAICS Sector	Sample 1	Sample 2	Sample 1	Sample 2
Panel A: Semiconductor and Construction Sectors				
Semiconductor Mfg (334413)	0.105	−0.085	−0.823	−0.311
	(0.202)	(0.200)	(0.979)	(0.606)
Nonres. Construction (2362)	−0.077	−0.105	0.968*	2.362*
	(0.054)	(0.068)	(0.430)	(1.096)
Building Equipment (2382)	−0.026	−0.043	0.341*	1.038*
	(0.023)	(0.034)	(0.154)	(0.433)
Panel B: Supply Chain Sectors				
Basic Chemicals (3251)	−0.151	−0.348	0.836**	2.500**
	(0.119)	(0.351)	(0.301)	(0.799)
Other Chemicals (3259)	0.015	0.154	1.192	−0.132
	(0.206)	(0.155)	(0.925)	(0.842)
Industrial Machinery (3332)	−0.144	−0.292	1.279*	1.768*
	(0.133)	(0.168)	(0.541)	(0.785)
Struct. Metals (3323)	−0.008	0.009	0.456	0.947
	(0.041)	(0.087)	(0.291)	(0.806)
Prof. Equipment Whls (4234)	−0.038	0.020	−0.242	−0.880
	(0.042)	(0.090)	(0.320)	(0.643)
Machinery Whls (4238)	−0.048*	−0.120**	0.521*	0.979
	(0.022)	(0.037)	(0.204)	(0.551)
Chemical Whls (4246)	−0.061	−0.017	−0.181	0.345
	(0.041)	(0.049)	(0.353)	(0.616)
Arch. & Engineering (5413)	0.000	0.009	0.310	0.508
	(0.028)	(0.039)	(0.185)	(0.547)
Panel C: Selected Two-Digit NAICS Service Sectors				
Wholesale Trade (42)	−0.028**	−0.043***	0.095	0.091
	(0.011)	(0.012)	(0.080)	(0.181)
Retail Trade (44-45)	−0.082	−0.199	0.321	0.779
	(0.065)	(0.117)	(0.172)	(0.552)
Information (51)	−0.183***	−0.180**	1.180***	1.707***
	(0.033)	(0.057)	(0.239)	(0.480)
Finance (52)	−0.050*	−0.068	0.247	0.371
	(0.021)	(0.044)	(0.137)	(0.302)
Professional Services (54)	−0.085***	−0.121***	0.235	0.033
	(0.016)	(0.036)	(0.123)	(0.341)
Accommodation & Food (72)	0.164***	0.061	−1.213***	−1.163**
	(0.046)	(0.060)	(0.205)	(0.418)
Other Services (81)	−0.050	−0.047	−0.401**	−0.942***
	(0.030)	(0.032)	(0.126)	(0.268)

Note(s): This table reports coefficients on the interaction term between post-treatment period (2021:Q2 onward) and the confounding variable, with standard errors in parentheses. The confounder is either the average remote-work adoption rate (columns 1–2) or the change in the county's average net population growth rate between the 2015–2020 and 2021–2025 periods (columns 3–4). Sample 1 includes counties with average 2015–2019 population $\geq$ 200,000; Sample 2 includes the 40 counties closest to Maricopa County by log population distance. All regressions include county and time fixed effects, are weighted by pre-2020 average sectoral employment, and cluster standard errors at the county level. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Source(s): Quarterly Census of Employment and Wages (QCEW); Current Population Survey (CPS); U.S. Census Bureau County Population Estimates; and authors' calculations.

that remote work restrained employment growth in this sector rather than raising it, biasing our estimated treatment effect downward. Differential remote-work adoption is therefore unlikely to account for the effects we estimate.

By contrast, several broader two-digit NAICS sectors show significant negative associations between remote work and employment growth, particularly information (NAICS 51), professional services (NAICS 54), and finance (NAICS 52). For these sectors, Maricopa County's above-average remote-work adoption may have constrained local employment growth independent of semiconductor investment. We therefore interpret the employment results for these sectors with caution and do not attribute the observed patterns solely to semiconductor spillover effects.

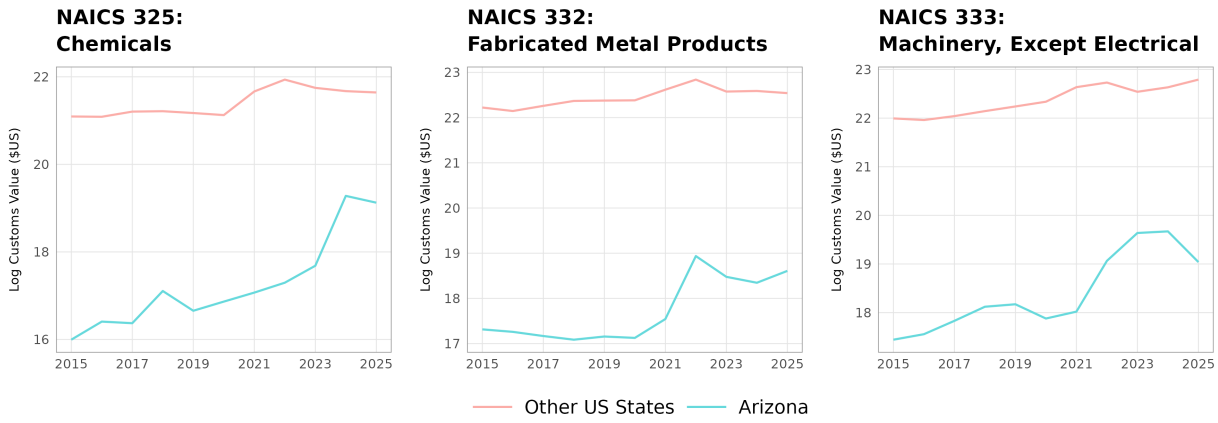
C.2 Population Growth and Employment Growth

The population growth results (columns 3–4) associate faster relative post-pandemic population growth with higher employment gains in construction (NAICS 2362, 2382), basic chemical manufacturing (NAICS 3251), industrial machinery (NAICS 3332), and information (NAICS 51) and lower gains in accommodation and food services (NAICS 72) and other services (NAICS 81). Maricopa County's net population growth slowed relative to its synthetic control over the post-treatment period, but the gap is small. An SDID analysis using the net population growth rate as the outcome returns an ATT of -0.35 percentage points in annual net population growth rate, which cannot be distinguished from zero.[11] The small gap suggests that population dynamics are unlikely to account for the employment effects we estimate, and in the positively associated construction and supply-chain sectors, the gap is expected to attenuate the observed employment effect instead of inflating it.

[11] The analysis uses the same donor pool construction as the employment estimates, so the ATT is Maricopa County's post-treatment deviation in the growth rate relative to its synthetic control. The estimate is -0.0035, not statistically significant under either placebo-based test.

D Arizona Imports from Taiwan

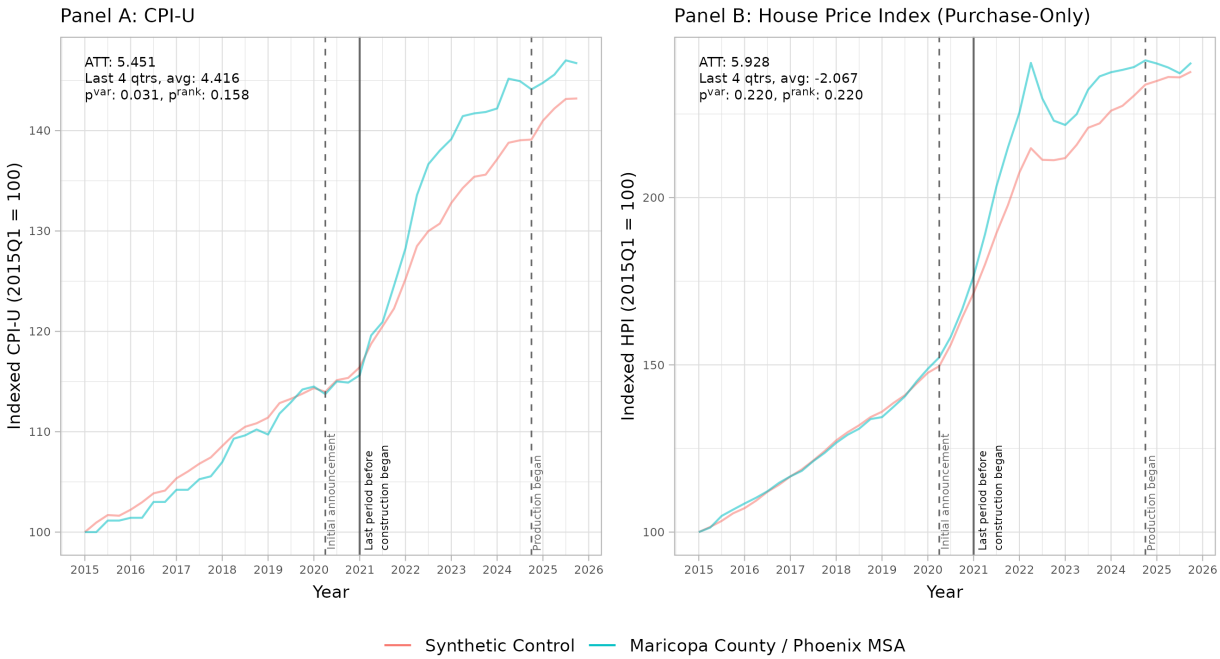
Figure A2: Arizona Imports from Taiwan by Manufacturing Category



Note(s): This figure shows Arizona's imports from Taiwan in three broad manufacturing categories—chemicals (NAICS 325), fabricated metal products (NAICS 332), and machinery (NAICS 333)—relative to other US states following the 2021 semiconductor investment.
Source(s): USA Trade Online and authors' calculations.

E Consumer Price and Housing Cost Dynamics

Figure A3: SDID Estimates for Local Price Indexes



Note(s): This figure shows the trajectories for the Consumer Price Index for All Urban Consumers (Panel A) and the FHFA Purchase-only House Price Index (Panel B) from 2015:Q1 to 2025:Q4 for Maricopa County or the Phoenix-Mesa-Scottsdale metropolitan statistical area (blue) and its synthetic control (red), constructed via the SDID method. The vertical lines indicate key events in the semiconductor investment timeline.
Source(s): US Bureau of Labor Statistics (BLS) Consumer Price Index and Federal Housing Finance Agency (FHFA) Purchase-only House Price Index for metropolitan areas, and authors' calculations.

F Spillovers to Neighboring Counties

Table A3: Employment Spillovers to Counties Surrounding Maricopa County

County and NAICS Sector	$\hat{\tau}$	p^{var}	p^{rank}
Pinal County			
Nonres. Construction (2362)	0.166	0.264	0.171
Building Equipment (2382)	0.493	<0.001	0.024
Architectural Metals (3323)	−0.485	0.988	0.976
Prof. Equip. Wholesale (4234)	0.548	0.037	0.098
Machinery Wholesale (4238)	0.302	0.067	0.049
Arch. & Engineering (5413)	0.127	0.274	0.268
Pima County			
Nonres. Construction (2362)	0.057	0.355	0.317
Building Equipment (2382)	0.022	0.348	0.317
Architectural Metals (3323)	−0.312	0.988	0.976
Semiconductor Mfg (3344)	−0.048	0.628	0.610
Prof. Equip. Wholesale (4234)	0.083	0.284	0.293
Machinery Wholesale (4238)	0.096	0.130	0.146
Chemical Wholesale (4246)	0.449	0.028	0.122
Arch. & Engineering (5413)	0.024	0.349	0.439
Yavapai County			
Nonres. Construction (2362)	0.140	0.327	0.268
Building Equipment (2382)	0.040	0.331	0.366
Architectural Metals (3323)	−0.080	0.645	0.683
Prof. Equip. Wholesale (4234)	−0.013	0.517	0.512
Machinery Wholesale (4238)	0.073	0.313	0.293
Arch. & Engineering (5413)	0.079	0.369	0.390
Yuma County			
Nonres. Construction (2362)	0.189	0.178	0.171
Building Equipment (2382)	0.005	0.482	0.439
Architectural Metals (3323)	0.062	0.403	0.512
Machinery Wholesale (4238)	−0.002	0.505	0.463
Arch. & Engineering (5413)	−0.023	0.569	0.561

Note(s): Each row reports $\hat{\tau}$, the average effect on log employment over the post-treatment period, and the one-sided variance-based and rank-based p-values p^{var} and p^{rank} against $H_1 : \tau > 0$. Estimation and inference follow Section 2, taking the county named as the treated unit. Every county adjacent to Maricopa County other than the one being examined is excluded from the donor pool. Some county-sector pairs are omitted because the pair fails the suppression and sector size restrictions of Section 2.
Source(s): QCEW and authors' calculations.

Table A3 reports SDID estimates for the four Arizona counties bordering Maricopa that are large enough to estimate: Pinal, Pima, Yavapai, and Yuma. We treat each in turn as the treated unit, holding the estimator, the treatment date of 2021:Q2, the donor pool selection criteria, and the placebo-based inference procedure of Section 2 fixed, and drawing 40 donors for every cell. We examine the same construction and supply chain sectors as the main analysis, substituting semiconductor manufacturing (NAICS 3344) for NAICS 334413, which no neighboring county reports. Coverage is limited by QCEW disclosure suppression, and 25 of the 44 county-by-sector cells are estimable. Basic chemicals (NAICS 3251), other chemical products (NAICS 3259), and

industrial machinery (NAICS 3332) are not estimable in any of the four counties.

The estimates give little indication of spillovers to neighboring counties. Only one of the 25 cells is distinguishable from zero at the 5 percent level under both one-sided tests: building equipment contractors (NAICS 2382) in Pinal County, at 0.493 log points, with p^{rank} at its floor of 0.024. Three wholesale trade estimates are positive and sizable and each passes one of the two tests: professional equipment (NAICS 4234) in Pinal County at 0.548 ($p^{\text{var}} = 0.037$, $p^{\text{rank}} = 0.098$), machinery (NAICS 4238) in Pinal County at 0.302 ($p^{\text{var}} = 0.067$, $p^{\text{rank}} = 0.049$), and chemicals (NAICS 4246) in Pima County at 0.449 ($p^{\text{var}} = 0.028$, $p^{\text{rank}} = 0.122$). The two largest negative estimates, in architectural and structural metals (NAICS 3323) in Pinal and Pima Counties at -0.485 and -0.312, fall below all but one of their 40 placebo estimates.[12] The remaining estimates are small or imprecise.

[12] Because both tests are one-sided against $H_1 : \tau > 0$, their p-values near one for these cells indicate only the absence of a positive effect and do not test whether the declines are distinguishable from zero.